

Problem-Solution Fit

Date	27 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • Farmers (small, medium, and large-scale) • Agricultural Researchers • Government Agriculture Departments • Agricultural Extension Officers • Policymakers	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Limited financial resources. • Lack of technical knowledge. • Limited access to smart devices. • Poor internet connectivity. • Limited access to soil testing facilities.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Available Solutions • Traditional farming practices • Soil testing laboratories • Agricultural experts • Government advisory services Pros • Reliable expert guidance • Soil analysis available • Practical farming experience Cons • Time-consuming • Limited accessibility • Generic recommendations • Often not personalized
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2. PROBLEMS / PAINS + ITS FREQUENCY [PR]

- Difficulty selecting the most suitable crop.
- Low crop yield due to poor soil analysis.
- Inefficient use of fertilizers and water.
- Crop losses caused by unsuitable environmental conditions.
- Frequent incorrect farming decisions due to lack of data.

9. PROBLEM ROOT / CAUSE [RC]

- Lack of accurate soil and environmental analysis.
- Limited awareness of suitable crops.
- Climate variability affecting productivity.
- Absence of intelligent decision-support systems.
- Inefficient resource management.

7. BEHAVIOR + ITS INTENSITY [BE]

- Depend on traditional farming methods.
- Seek advice from local experts.
- Occasionally use online agricultural resources.
- Make decisions based on past experience.

3. TRIGGERS TO ACT [TR]

- Declining agricultural productivity.
- Increasing cultivation costs.
- Need for higher crop yield.
- Demand for sustainable farming.
- Requirement for data-driven agricultural decisions.

10. YOUR SOLUTION [SL]

OptiCrop is an AI and Machine Learning-based agricultural production optimization system that analyzes soil nutrients (Nitrogen, Phosphorous, Potassium), temperature, humidity, pH, and rainfall to recommend the most suitable crop. The system helps farmers improve crop yield, optimize resource utilization, reduce cultivation risks, and support sustainable farming through data-driven recommendations.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- Agricultural websites
- Mobile applications
- YouTube
- WhatsApp farmer groups

4. EMOTIONS BEFORE / AFTER [EM]

Before

- Confused
- Worried
- Uncertain
- Frustrated

After

- Confident
- Informed
- Satisfied
- Optimistic

OFFLINE

- Agricultural extension officers
- Local farming communities
- Soil testing centers
- Agricultural training programs